

Radar Histogram Analysis & Gain Selection

Overview of Coverage, Entropy & SNR Metrics

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Goals/Metrics Overview

- **Goal:** Choose a few (3-5) gain(s) that maximize dynamic range without drowning in noise. This is used for HDR in future
- What we will look at for this
 - Coverage: how many brightness levels are seen
 - Entropy: how evenly those levels are used
 - SNR: signal strength vs. noise floor

Coverage - an overview

- **Definition:** number of non-empty histogram bins
- **Formula:** $\text{coverage} = \sum_k [\text{hist}[k] > 0]$
- Important so we don't miss bright or faint echoes
- What data means:
 - Rising vs. gain -> capturing more dynamic range (good stuff)
 - Flat/falling -> saturation or clipping (not good usually)

Entropy, Shannon's Version

- **Definition:** How evenly or equally all histogram bins are filled
- Contrast with coverage:
 - Coverage = “how many” bins
 - Entropy = “how evenly” samples fill them
- Important to find the gains that uses ADC codes most efficiently
- How to analyze:
 - Peak indicates best uniformity
 - Low at extremes (all in one bin or stuck at noise floor)

SNR – Signal to Noise Ratio

- We need to find noise first
 - Baseline, assume bottom 5% is noise(cut out [0, MAX] from data)
 - Primary: Differential Noise Prediction (bs name I came up with)
 - Use differences(diffs) between sweeps, pulses, or cells/bins to calculate noise
 - Sweeps = slow; pulses = med; cells/bins = fast
 - Find center/median of ALL diffs
 - Calculate Scatter or (MAD) Median Absolute Deviation
 - Discard anything 3 MADs away from median(outliers)
 - Convert to std dev by taking std dev of the diffs remaining, then divide by $\sqrt{2}$
 - Give you new noise level. If this data is bad, revert to baseline noise
- Use $SNR = \text{signal_mean} / \text{std dev}$

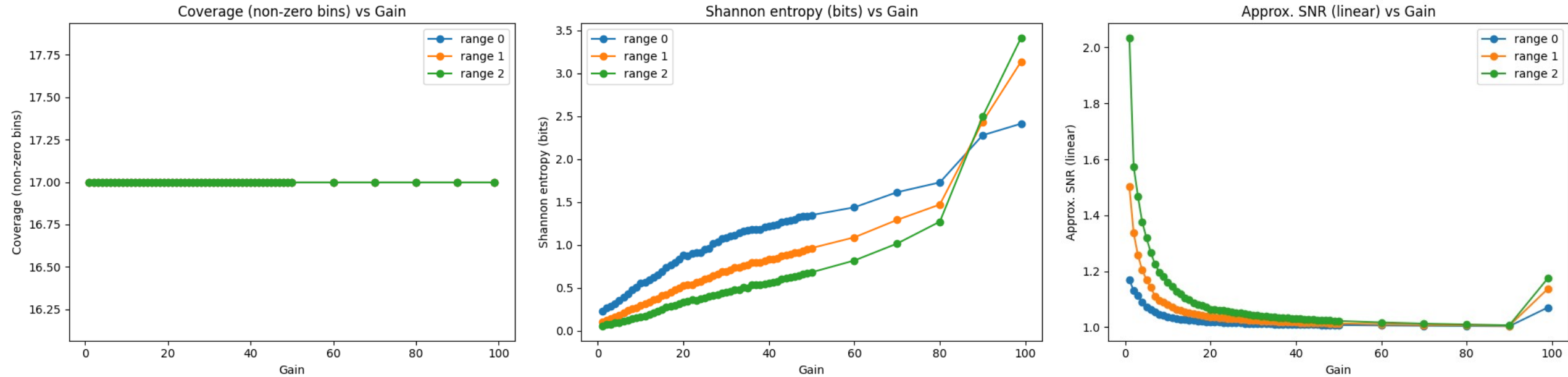
Summary

High coverage → see full dynamic range

Peak entropy → use ADC evenly

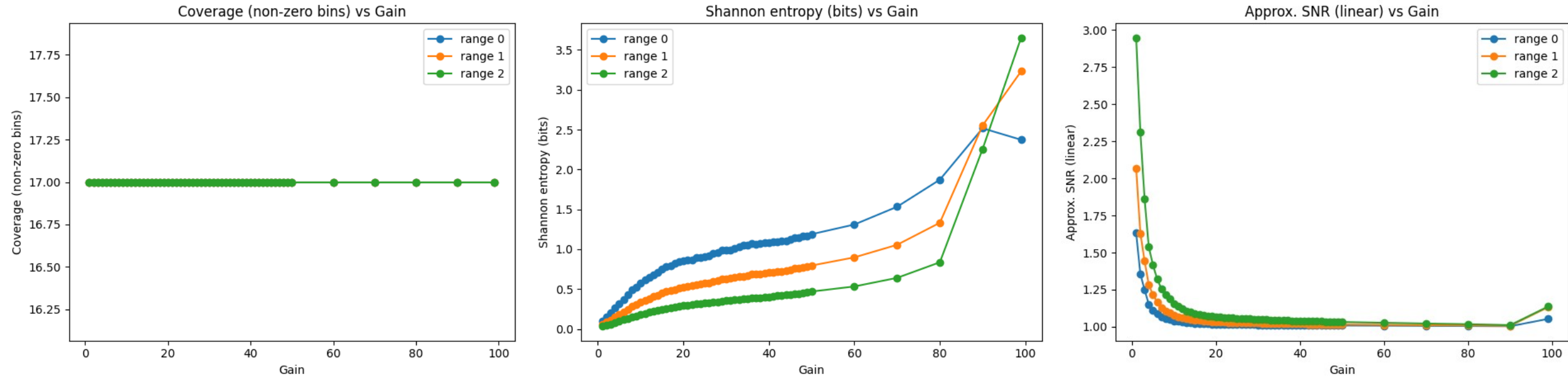
Acceptable SNR → keep noise under control

Dataset 1 results



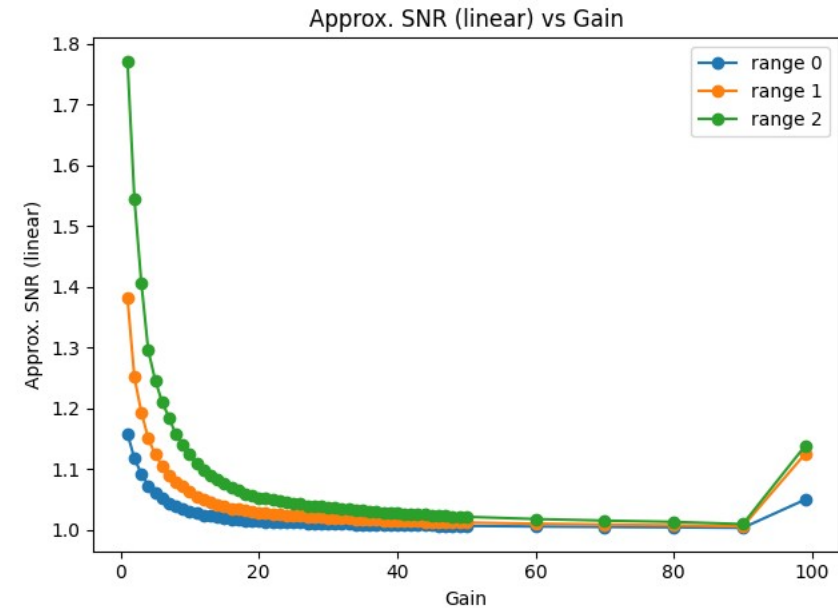
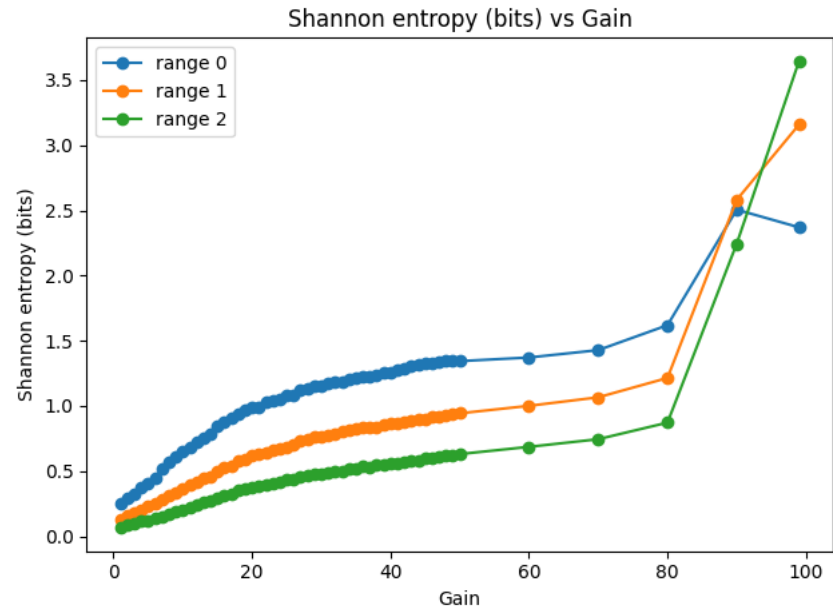
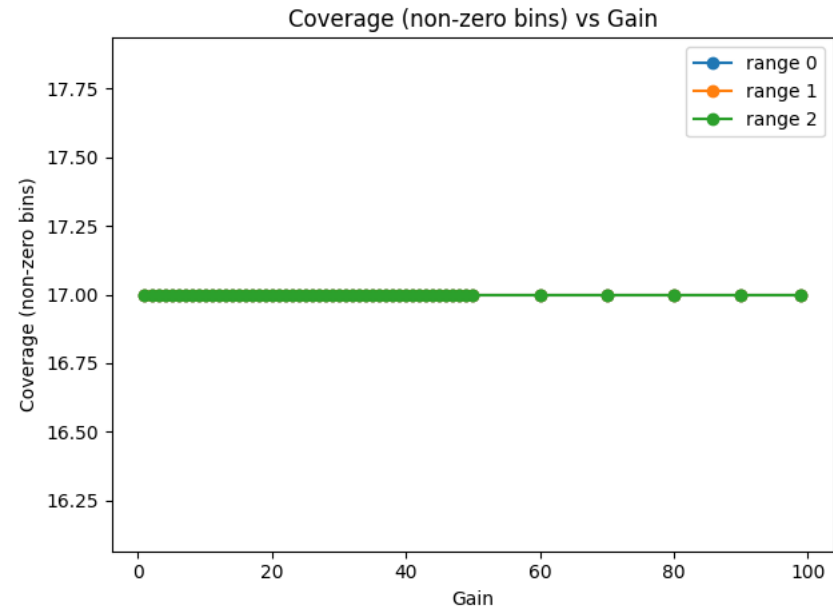
- **Notes:** Night time data; higher wave activity

Dataset 2 results



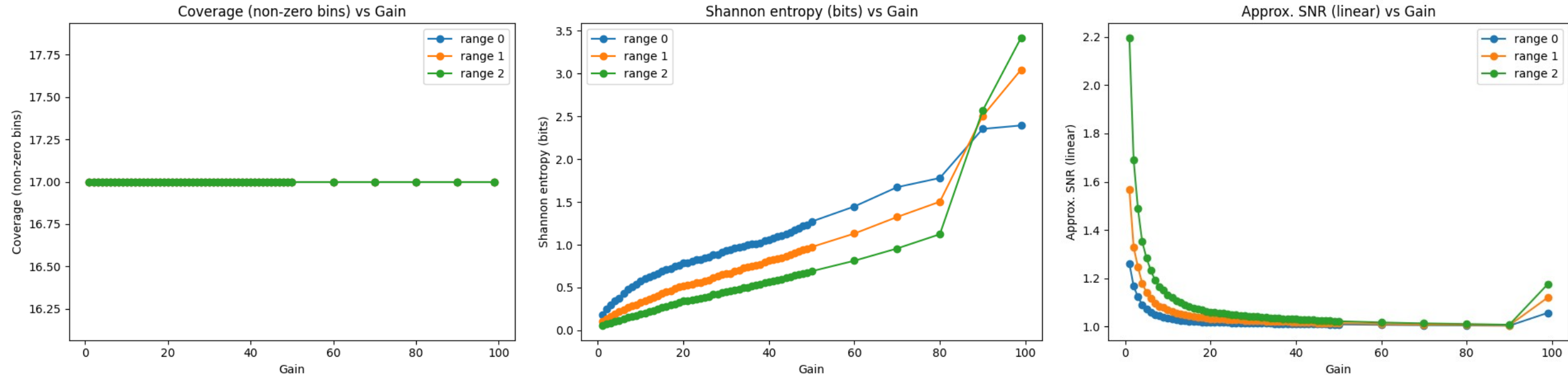
- **Notes:** interpolation off

Dataset 3 results



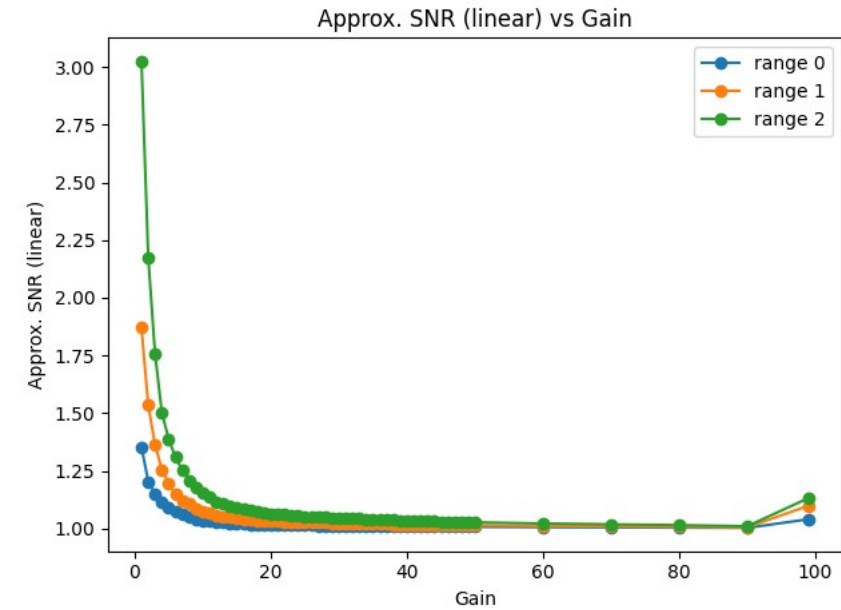
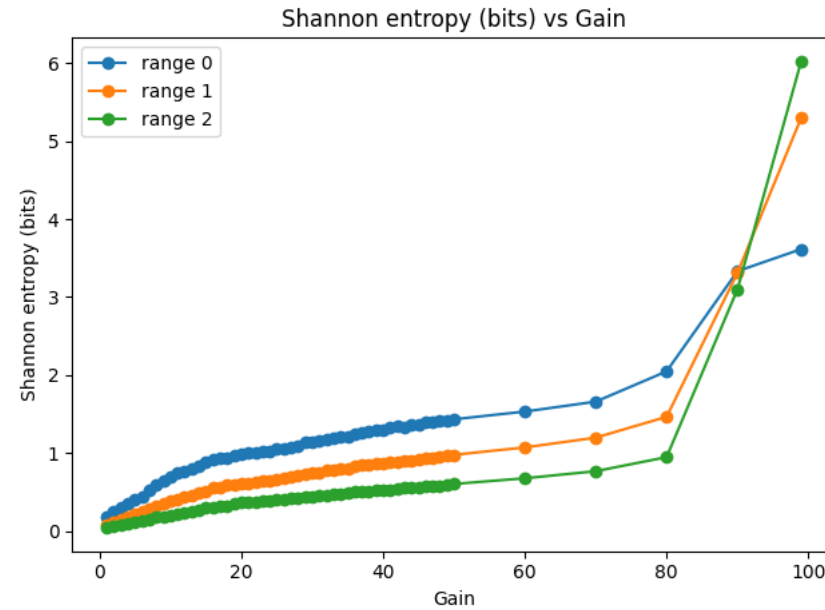
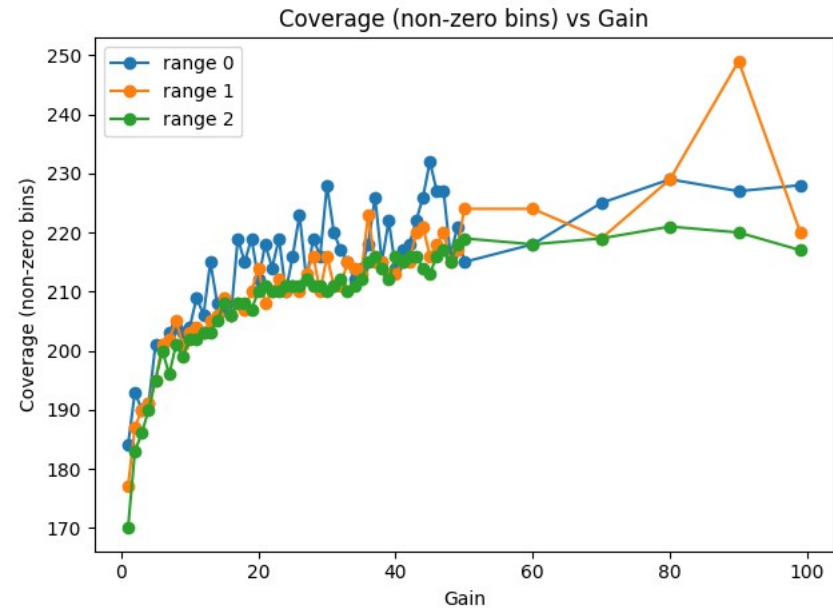
- **Notes:** interpolation off

Dataset 4 results



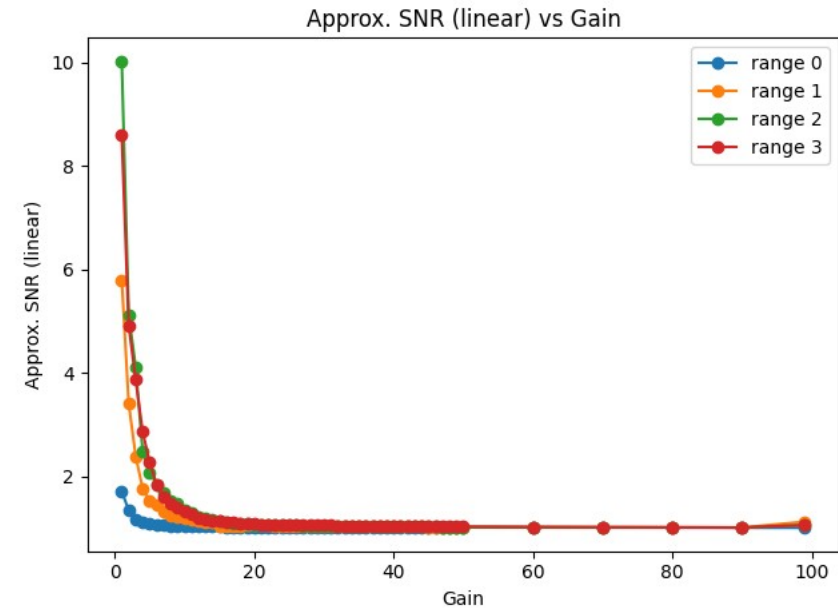
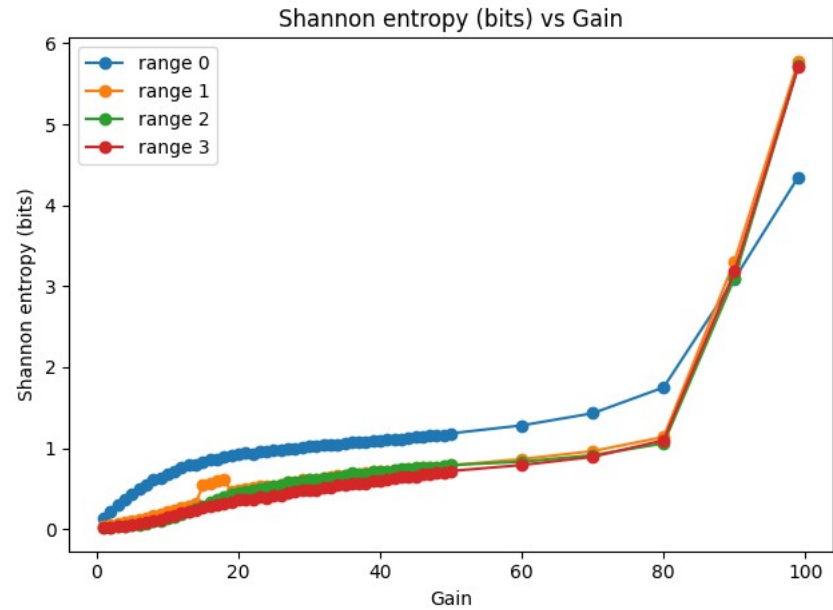
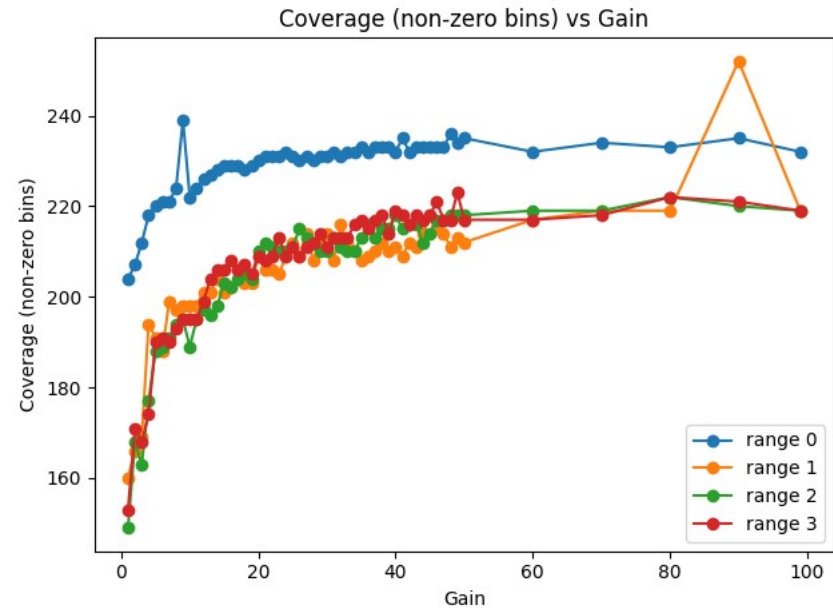
- **Notes:** interpolation off

Dataset 5 results



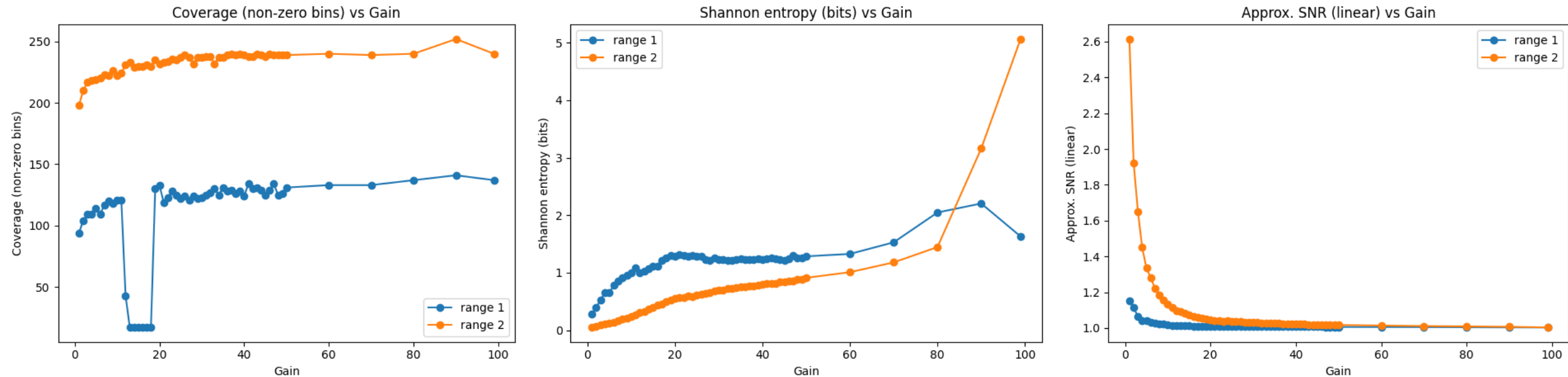
• **Notes:**

Dataset 6 results



- **Notes:**

Dataset 7 results



- **Top 5 by entropy:** R1: 90,80,99,11,70 || R2: 99,90,80,70,60
- **Top 5 by k-means:** R1: 27,2,1,4,8 || R2: 25,3,7,8,9
- **Notes:** Testing data for STC & Pulse Width. Not for calibration

CODE

Main python file^^^

Helper python file^^^